Oct 10, 2025

AULE SPACE INC.
Jetpacks for satellite life-extension

GNC Technical Assignment

Assignment Tasks

1. Implement a 3 DOF spacecraft attitude kinematics and dynamics block in Simulink using quaternion representation.

2. Design and implement an optimal (non-PID) quaternion-based control law in Simulink to perform spacecraft attitude control.

Simulation Parameters

Parameter	**Value / Scenario**
Initial Attitude (Euler angles)	[12°, 31°, -40°]
Final Desired Attitude (Euler angles)	[40°, -30°, 90°]
Initial Body Rates (deg/s)	[10, -20, 12]
Final Desired Body Rates (deg/s)	[0.0, 0.0, 0.0]
Satellite Moment of Inertia ($kg \cdot m^2$)	[[1.10, 0.05, 0.00], [0.05, 1.90, -0.01], [0.00, -0.01, 1.12]]
Reaction Wheel Configuration	3 Reaction wheels
Disturbance Torque (ECI frame)	Sinusoidal, 1×10^{-6} Nm

Note: Assume and specify reasonable values for parameters wherever required.

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Theoretical Question

In the closed-loop simulation of the satellite AOCS, an attitude estimation block is essential for robust performance. Explain how you would design and implement an Extended Kalman Filter (EKF) for spacecraft attitude estimation using IMU (gyroscope and accelerometer) and Star Tracker data. Discuss:

• How does the EKF fuse IMU and Star Tracker measurements to estimate spacecraft attitude?

• How do estimation errors impact the performance of the control law?

• Include your philosophy on initializing and tuning filter noise covariances.

Expected Deliverables

- Complete the provided Simulink model with system components and control law.
- Plots for satellite body torques, attitude, body rates, and wheel momentum showing convergence and performance.
- A report including method of tuning control parameters, results with interpretation (overshoot, settling time, control effort) and discussion of EKF theory.

Submission Guidelines

Submit all deliverables, including Simulink files and the performance report, via the following Google Form.

If you encounter any issues or have questions regarding the challenge, feel free to reach out to admin@aule.space.